Setup (for MacOS and Linux)

1. Create a virtual environment:

   $ python3 -m venv .venv

2. Activate the virtual environment:

   $ source .venv/bin/activate

3. Install the following Python libraries:

   - **pandas**
   - **openpyxl**

   (.venv) $ pip3 install pandas openpyxl

4. Create four folders in the root directory: **feedback_1, feedback_2, feedback_3, combined_datasets**

5. Ensure all the spreadsheets in the ***feedback*** folder are sorted by name (alphabetical order).

   Then take the first 20 spreadsheets in the feedback folder (*Ballarat – Customer Satisfaction.xlsx to Hunter Motors feedback 25-26.xlsx*) and transfer them to the ***feedback_1*** folder.

   Do the same thing with the next 20 (*IPS_feedback_log.xlsx to Penrith Feedback.xlsx*) to the ***feedback_2*** folder.

   Finally, move the remaining 12 spreadsheets (*Ringwood feedback Q1 2026.xlsx to Wollongong_Oct25-March26.csv*) into the ***feedback_3*** folder.

Running scripts to get the single combined Dataset

1. Run the ***merge_feedback_1.py*** script to combine the first 20 spreadsheets into a single spreadsheet called ***merged_feedback_1.csv*** in the ***combined_datasets*** folder.

   (.venv) $ python3 merge_feedback_1.py

2. Run the ***merge_feedback_2.py*** script to combine the next 20 spreadsheets into a single spreadsheet called ***merged_feedback_2.csv*** in the ***combined_datasets*** folder.

   (.venv) $ python3 merge_feedback_2.py

3. Run the ***merge_feedback_3.py*** script to combine the last 12 spreadsheets into a single spreadsheet called ***merged_feedback_3.csv*** in the ***combined_datasets*** folder.

(.venv) $ python3 merge_feedback_3.py

4. Run the ***merge_all_feedback.py*** script to combine the three ***merge_feedback_{1, 2, 3}.csv*** files together into a single spreadsheet called ***merged_all_feedback.csv*** in the ***combined_datasets*** folder.

   (.venv) $ python3 merge_fall_feedback.py

5. Run the ***clean_merge_all_feedback.py*** script to perform all the cleaning processes on the ***merged_all_feedback.csv*** file to get a single spreadsheet called ***merged_all_feedback_cleaned.csv*** in the ***combined_datasets*** folder.

   (.venv) $ python3 clean_merge_all_feedback.py

6. Manual Intervention requirement:
   Open ***merged_all_feedback_cleaned.csv*** in Excel and select the *Filter/AutoFilter* option to provide a drop down menu for the column names on the first row.
   For 'Suburb', filter only for 'Tamworth'. With the filtered rows, look for the 2 rows that have an empty Rating/Score AND Comment entries then delete the rows they are on.

Final Single Combined Dataset: *combined_datasets/merged_all_feedback_cleaned.csv* (with 7041 rows by 5 columns)

For Sentiment Analysis

- Run the ***sentiment_analysis.py*** script to add a new column called '*Sentiment from Comment*' to the ***merged_all_feedback_cleaned.csv*** file to get an updated dataset called ***merged_all_feedback_cleaned_withSentAnaly.csv*** in the ***combined_datasets*** folder.

  (.venv) $ python3 sentiment_analysis.py

  Note: This script takes longer to run